YUXI ZHAI

Shanghai, China | 386079267@qq.com
zedkyx.github.io | github.com/zedkyx

EDUCATION

Shanghai University, School of Mechanics and Engineering Science **2025–2028 (expected)**
Master's student in Mechanics; research focus: Fluid Mechanics | GPA: 3.5/4.0

Shanghai University, Qian Weichang College **2021–2025**
B.Sc. in Theoretical and Applied Mechanics

RESEARCH EXPERIENCE

Neural Operators for Variable-Geometry Flows **May 2026–Present**

- Adapted geometries in an existing FEniCSx solver to generate 2D obstacle-flow datasets and trained Transolver predictors.
- Applied a discrete weak-divergence projection to reduce velocity divergence errors; evaluated predictions on unseen geometries.

Nodal-Measure Weighting for Stress Prediction **May 2026–Present**

- Developed finite-element nodal weighting for Transolver, improving area-weighted stress prediction on irregular meshes.
- Extended the shell-joint study from COIN's 4 geometric parameters to 25 geometric parameters and 1 loading parameter, moving from displacement to stress prediction.

First-author manuscript in preparation.

COIN: Component-Aware Operator Learning for Shell Design **Feb 2025–May 2026**

- Developed separate networks for radial and axial displacement prediction, reducing radial relative L2 error by approximately 30% on average across three neural-operator architectures against their shared-network baselines.
- Combined geometry-conditioned predictions with a genetic algorithm to minimize structural volume under a displacement constraint.

First-author manuscript being revised for resubmission.

PINNs for Steady Lid-Driven Cavity Flow **Dec 2023–Oct 2024**

- Implemented PINNs for steady cavity flow at Re = 100 and 1,000; investigated entropy-viscosity stabilization and compared velocity profiles with benchmarks.
- Extended the undergraduate research to FNO prediction of transient Stokes and cavity flows (Apr–May 2025).

HONORS AND AWARDS

First-Class Graduate Academic Scholarship, Shanghai University 2025
First Prize, Undergraduate Innovation Project 2024
Shanghai Frontier Science Research Base of Mechanics Informatics, Shanghai University
Outstanding Participant, Mechanics Summer School, Zhejiang University 2024
Second-Class Academic Excellence Scholarship, Shanghai University 2023–2024

SKILLS AND ADDITIONAL INFORMATION

Programming & methods: Python, PyTorch; PINNs, FNO, DeepONet, Transolver.
Simulation software: FEniCSx, Abaqus.
English: IELTS Academic 8.0.
Interests: Badminton — singles quarterfinalist in a university tournament.